

Practice the measurements

Name _____

First 10 minutes. Each problem is separate. Show the formula, substitute the values, and carry the units through your calculation.

Formula box

$$X = C / T$$

throughput

$$S = B / C$$

service time

$$U = B / T = X \cdot S$$

Utilization Law

$$N = X \cdot R$$

Little's Law

$$R = N / X$$

latency from Little's Law

$$\text{wait} = R - S$$

time inside but not in service

1. **Throughput, X.** A service completes **C = 24 jobs** during **T = 30 minutes**. Calculate X in jobs per minute and jobs per hour.

2. **Service time, S.** A printer records **B = 42 busy minutes** while completing **C = 7 jobs**. Calculate S.

3. **Utilization, U.** During **T = 60 minutes**, a disk is busy for **B = 48 minutes** and completes **C = 8 requests**. Calculate U using B/T. Then calculate X and S and verify the same U using X · S.

4. **Latency, R.** A system has an average of **N = 2 jobs inside** and throughput **X = 8 jobs/hour**. Use Little's Law to calculate R in hours and minutes.

5. **Waiting.** For the system above, **R = 15 minutes** and **S = 5 minutes**. Calculate average waiting time.

6. **Why latency is not 1/X.** If **X = 8 jobs/hour**, calculate 1/X. Then explain why that result is not the 15-minute latency when **N = 2 jobs** are inside on average.

Put it together: assignment-upload system

Name _____

Next 10 minutes. Use one measured system record to calculate X , S , U , R , and waiting time. Finish by deciding which part of the system deserves attention.

Boundary: One upload request enters when the service accepts it and leaves when confirmation returns. The boundary contains one upload processor and its waiting line.

Measured over $T = 20$ minutes	Symbol and value
Completed uploads	$C = 8$ uploads
Total time the processor was busy	$B = 16$ minutes
Average uploads inside the boundary	$N = 1.2$ uploads

1. Calculate throughput $X = C/T$ in uploads per minute.

2. Calculate service time $S = B/C$ in minutes per upload.

3. Calculate utilization directly using $U = B/T$.

4. Check the Utilization Law: substitute your X and S into $U = X \cdot S$. Show how the units cancel.

5. Use Little's Law to calculate latency: $R = N/X$.

6. Calculate the average wait: **wait** = $R - S$. Then label the latency bar.

[waiting: _____ min] [service: _____ min] = R : _____ min

7. **Faster processor or another processor?** Which should the team investigate first if its goal is lower latency? Use your service and waiting times as evidence. State what each option directly attacks.

Decision rule: Find where the time goes. A faster resource attacks service time; another resource attacks waiting caused by requests competing for service. These measurements support what to investigate next—not an unmeasured performance promise.